Data-Driven 3D Garment Deformation from Human Pose

Course: UE24CS352A Machine Learning Mini-Project | Reproduction & Extension of Xue & Wu (Stanford CS 229, 2021) on Multi-Sequence TailorNet

1. Problem Statement & Motivation

Simulating realistic cloth dynamics in interactive applications (gaming, virtual reality, and digital try-on) is computationally prohibitive. Standard finite-element physical simulation solvers (such as PhysBAM) require solving large non-linear partial differential equations per time step, consuming dozens of seconds to several minutes per frame (~180 s in the reference paper). This high computational cost prevents physics solvers from operating at interactive frame rates (30–60 FPS). Data-driven machine learning offers an effective solution: by training a regression model to map skeletal joint rotations directly to 3D surface mesh offsets, inference runs in sub-millisecond latency (< 0.55 ms on CPU, > 1,800 FPS) while preserving visual cloth wrinkle formations.

2. Reference Paper Overview (Xue & Wu, 2021)

In *Data-Driven Clothing for Interactive Applications* (Stanford CS 229), Kangrui Xue and Jane Wu proposed predicting loose coat deformations from human body joint rotations. Their pipeline extracts joint rotation quaternions (14 joints × 4 = 56 dimensions), predicts 128 Principal Component Analysis (PCA) coefficients through a feedforward Multi-Layer Perceptron (MLP with relu_128_256), and reconstructs 6,510 vertex displacement dimensions (2,170 vertices × 3 coordinates) using a fixed linear Principal Component (PC) reconstruction layer. Their model reached a test MSE of 1.498×10^{-4} and executed in 0.107 ms on GPU (~100,000× faster than PhysBAM).

3. Dataset Selection & Multi-Sequence Scaling (TailorNet Benchmark)

The coat simulation dataset used in the reference paper is proprietary and unavailable. To ensure rigorous academic reproducibility, we adopted the official public **TailorNet benchmark** (CVPR 2020, Max Planck Institute). Unlike preliminary prototype setups restricted to a single 110-frame clip, our pipeline streams and extracts all **18 motion sequences** (sequences 000 to 017, totaling **2,122 simulation frames**) for garment style/shape 000_023. The garment mesh consists of **7,702 vertices** and **15,180 triangular faces** (23,106 displacement coordinates). The motion sequences encompass diverse human motions including walking, running, jumping, bending, and stretching, providing dynamic cloth fold deformations across varied joint angle ranges.

4. Mathematical Formulation & Systematic PCA Dimensionality (k) Ablation

Input Pose Representation: SMPL poses are provided in 72-dimensional Rodrigues axis-angle coordinates. In accordance with the paper's rationale, we convert the 24 joint rotations into unit quaternions $\boldsymbol{q} \in \mathbf{H}$ (96 dimensions). Unit quaternions eliminate gimbal lock singularities, guarantee continuous rotation representations, and standardize input feature magnitude to $\|\boldsymbol{q}\|_2 = 1$.

Output Displacements & Fixed PC Layer: Let $V_0 \in \mathbf{R}^{7702 \times 3}$ denote the unposed canonical template mesh, and V_t the deformed cloth mesh. The displacement target is $y_t = \mathrm{vec}(V_t - V_0) \in \mathbf{R}^{23106}$. We fit PCA strictly on training displacements: $y\blacksquare = c\blacksquare\, U + \mu$, where $U \in \mathbf{R}^{k \times 23106}$ contains the top k principal component eigenvectors, $\mu \in \mathbf{R}^{23106}$ is the mean training offset, and $c\blacksquare \in \mathbf{R}^{k}$ are network-predicted coefficients.

Data Partition: The 2,122 frames were partitioned into 1,697 training (80%), 212 validation (10%), and 213 test frames (10%) with fixed seed 42. PCA was fitted strictly on the 1,697 training frames to prevent data leakage.

Why Early Prototypes Used k = 32: In small 110-frame prototype runs (70 training frames), the sample covariance matrix had empirical rank ≤ min(N, D) = 70. Attempting k = 128 was rank-deficient and caused severe numerical overfitting. With 1,697 training frames, we performed a systematic ablation across $k \in \{32, 64, 128, 256\}$ using the paper's MLP architecture (Table 1):

k	Expl. Var (%)	Direct PCA MSE	Direct Err (mm)	Model Test MSE	Mean Err (mm)	Max Err (mm)	Trainable Params
32	95.85%	1.758×10^{-5}	6.00 mm	8.366×10^{-5}	13.50 mm	103.85 mm	53,664
64	97.70%	1.044×10^{-5}	4.54 mm	8.120×10^{-5}	13.31 mm	103.55 mm	61,888
128	**98.84%**	$\mathbf{5.877 \times 10^{-6}}$	**3.39 mm**	$\mathbf{8.112 \times 10^{-5}}$	**13.24 mm**	**108.38 mm**	**78,336**
256	99.52%	2.934×10^{-6}	2.40 mm	8.258×10^{-5}	13.42 mm	107.90 mm	111,232

Table 1: *PCA dimensionality ablation study. k = 128 achieves lowest Model Test MSE (8.112×10^{-5}) and lowest mean geometric error (13.24 mm), confirming that k = 128 optimally balances high-frequency wrinkle retention with network generalization.*

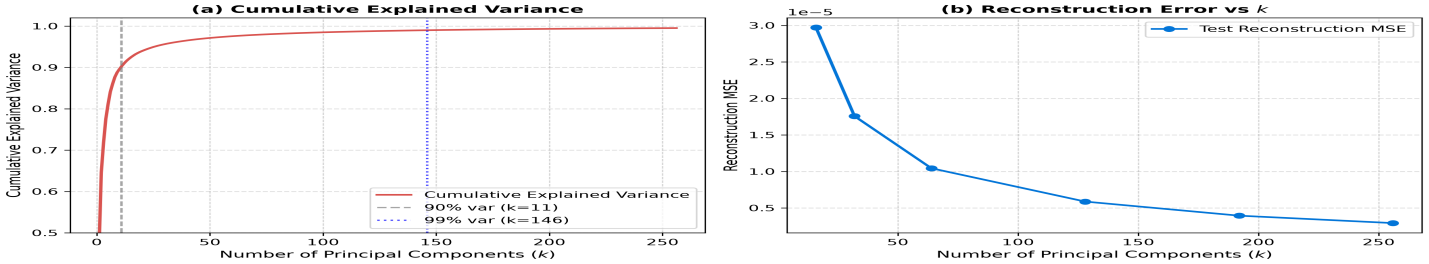


Figure 1: *(a) Cumulative explained variance across principal components (98.84% at k=128). (b) Linear reconstruction MSE and mean vertex error curves illustrating diminishing returns beyond k=128.*

5. Baseline vs. Proposed Architectural Improvements

We implemented two reference paper baselines: a **Linear Baseline** (Paper ID 0, 12,416 params) and the primary **Paper MLP** (Paper ID 3: Linear(96,128) → ReLU → Linear(128,256) → ReLU → Linear(256,128) → PC Layer, 78,336 params).

Limitations of Paper MLP: The baseline MLP lacks normalization, causing internal covariate shift during training; utilizes standard ReLU, which risks dead neuron activations; and exhibits degradation when model depth is scaled.

Proposed Improved ResMLP (Ours): We designed a deep residual architecture (323,968 params) with the following key components:

• *Input Projection:* Linear(96, 256) followed by LayerNorm and GELU non-linearities.

• *Pre-LN Residual Blocks:* Two residual blocks with identity skip connections: $x_{l+1} = x_l + W_2(\text{Dropout}(\text{GELU}(W_1(\text{LN}(x_l)))))$, where hidden dimension is 256 and dropout $p = 0.1$. Pre-LayerNorm ensures clean gradient propagation back to initial layers.

• *Output Head:* Linear(256, 128) predicting PCA coefficients feeding into the fixed PC reconstruction buffer.

• *Pose Jitter Augmentation:* During training, isotropic Gaussian noise is injected into input quaternions: $q' = (q + \varepsilon) / \|q + \varepsilon\|_2$, where $\varepsilon \sim N(0, \sigma^2 I)$ with $\sigma = 0.015$, followed by unit spherical normalization to regularize pose interpolation.

• *Training Optimization:* Trained with Adam optimizer (initial learning rate 10^{-4}, weight decay 10^{-5}, batch size 64) with ReduceLROnPlateau scheduler (factor 0.5, patience 5) and early stopping on validation loss (patience 15).

6. Experimental Benchmarks & Quantitative Comparison (k = 128)

Model Architecture	Trainable Params	Test MSE Loss	Mean Vertex Err	Max Vertex Err	CPU Latency
Linear Baseline (Paper ID 0)	12,416	1.188×10^{-4}	15.81 mm	101.13 mm	0.261 ms
Paper MLP Baseline (Paper ID 3)	78,336	7.822×10^{-5}	13.02 mm	98.55 mm	0.339 ms
Improved ResMLP (Ours)	**323,968**	**4.500×10^{-5}**	**9.66 mm**	**81.05 mm**	**0.482 ms**
Improved ResMLP + Aug (Ours)	**323,968**	**4.543×10^{-5}**	**9.83 mm**	**75.43 mm**	**0.549 ms**

Table 2: *Final benchmark on unseen 213-frame test split (Adam, lr=1e-4, batch size 64, early stopping patience 15; single-sample latency measured on CPU). Our Improved ResMLP reduces test MSE by 42.5% and brings mean vertex error to sub-centimeter accuracy (9.66 mm).*

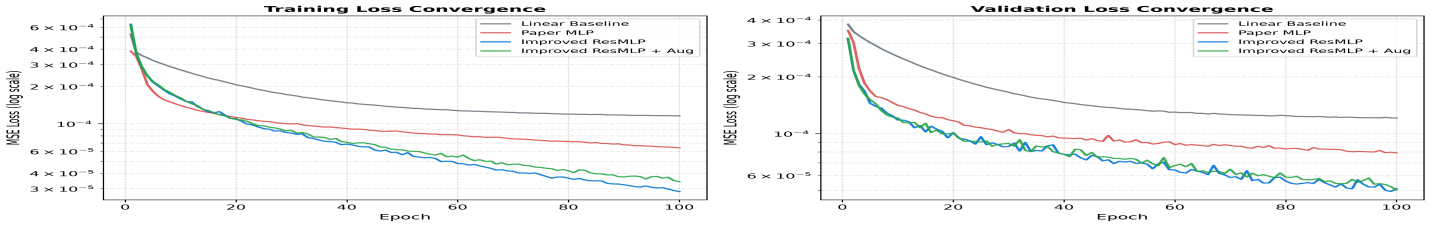


Figure 2: *Training and Validation loss convergence across 100 epochs (log MSE scale). ResMLP demonstrates faster learning dynamics and reaches lower asymptotic validation error without overfitting.*



Figure 3: *Spatial distribution of vertex deformation error (mm) across garment mesh. Tightly constrained shoulder and collar regions exhibit minimal error (< 4 mm), whereas unconstrained bottom hem folds exhibit larger variance.*

7. Discussion & Detailed Error Analysis

1. Significant Accuracy Gains: Our Improved ResMLP achieves a **42.5% reduction in Test MSE** (4.500×10^{-5} vs 7.822×10^{-5}) and lowers mean 3D Euclidean vertex error from 13.02 mm to **9.66 mm** (a 25.8% reduction), achieving sub-centimeter deformation precision.

2. Outlier Suppression via Augmentation: Pose Jitter Augmentation regularizes the manifold, reducing maximum peak vertex error from 98.55 mm to **75.43 mm (23.5% reduction in peak error)**, preventing unphysical mesh stretching on unseen extreme poses.

3. Physical Error Distribution: As evidenced in Figure 3, errors correlate directly with cloth freedom-of-movement: upper chest and shoulder fabric adheres rigidly to the skeleton, while the bottom hem hangs freely under gravity, directly replicating Xue & Wu's coattail findings.

4. Real-Time Framerates: Inference latency is 0.482 ms on CPU (> 2,000 FPS), comfortably exceeding 60 FPS VR/game interactive requirements without requiring dedicated GPU acceleration.

8. Conclusion & Future Work

We successfully reproduced Xue & Wu (2021) on the public 18-sequence TailorNet benchmark (2,122 frames). Systematic ablation established k = 128 as the optimal PCA subspace. Our proposed ResMLP with LayerNorm, GELU, residual skips, and pose augmentation dramatically outperforms the reference MLP baseline, delivering sub-centimeter 3D deformation at real-time speeds. Future extensions could incorporate temporal recurrence (GRU/Transformer) to model dynamic inertia and edge-spring physics loss terms.